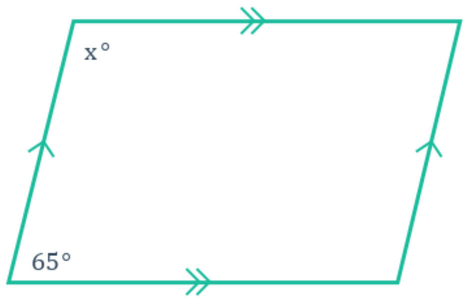
Properties of parallelograms

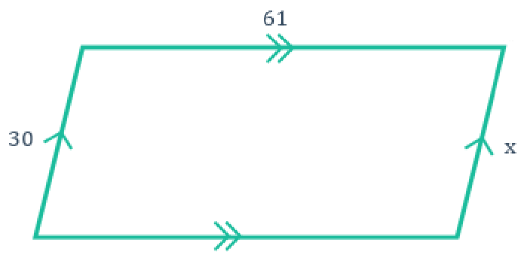
1 Find the value of x in each of the folowing parallelograms:

a

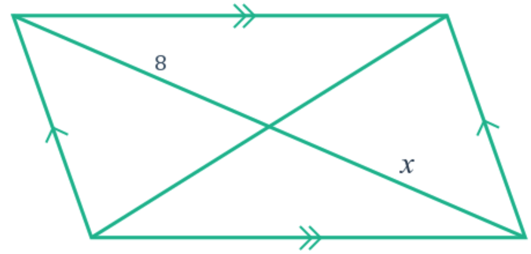


b

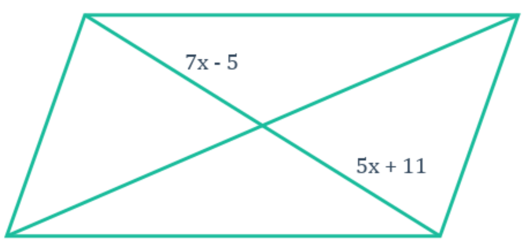
c



d

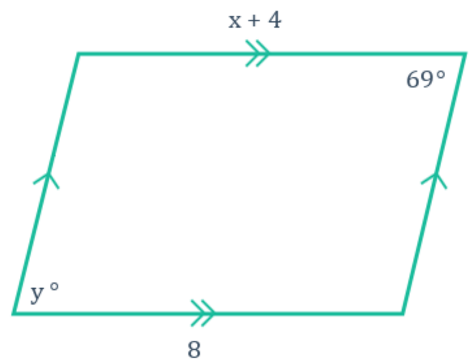


e

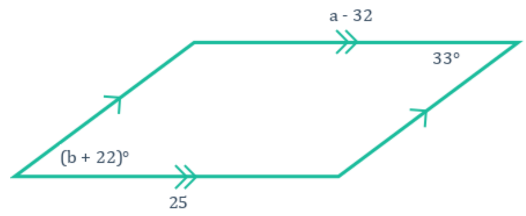


2 Find the value of the variables in the follow[redacted]agrams:

a

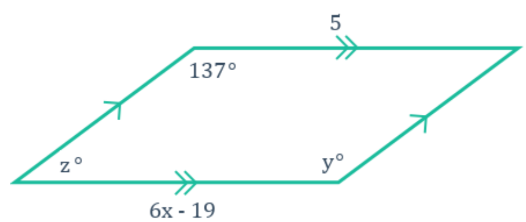


b

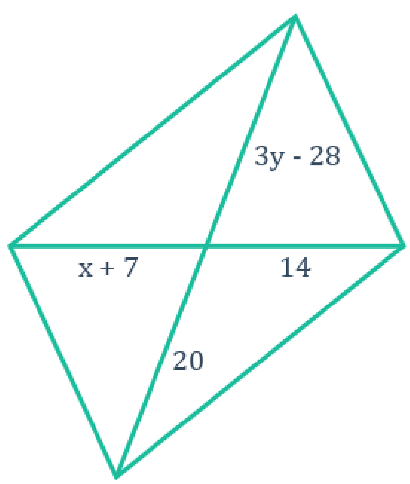


c

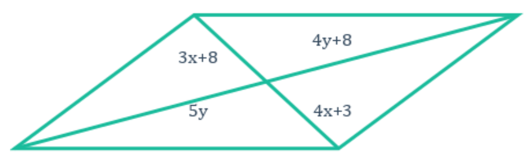
d



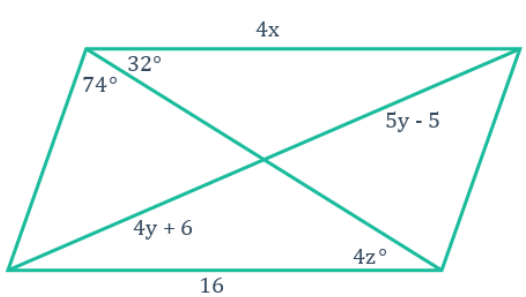
e



f

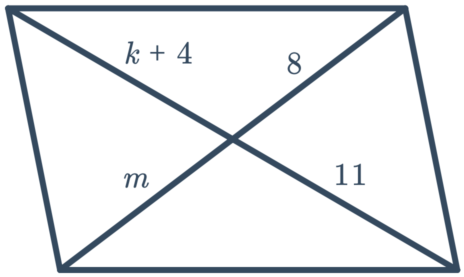


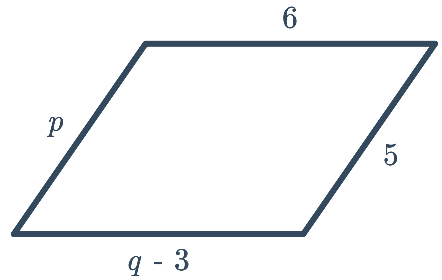
g



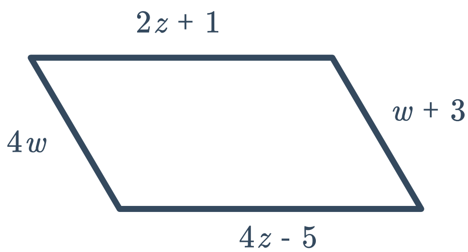
h

i

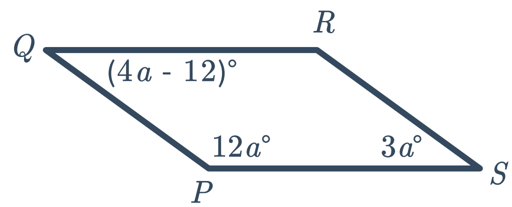




k



l

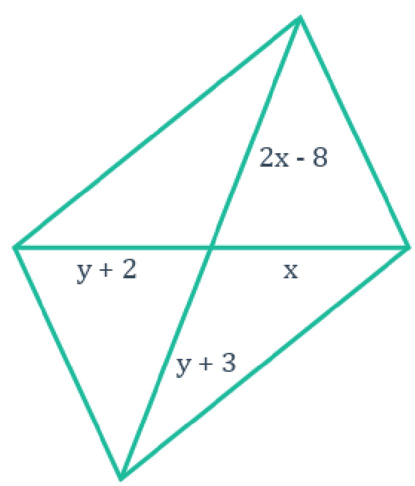


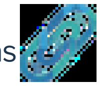
3 Find the value of x if the perimeter of the parallelogram given is 48 m.

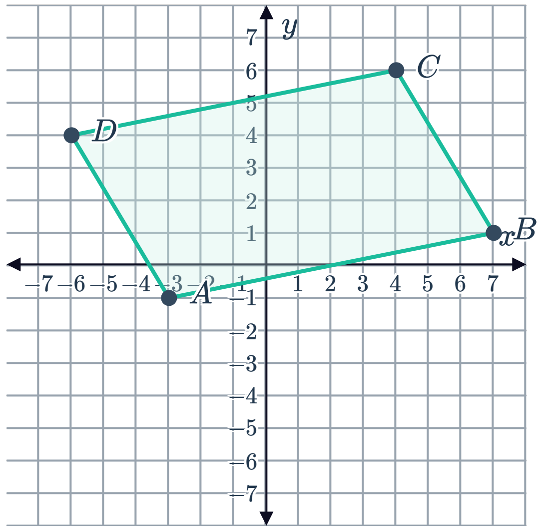
4 Consider the given parallelogram:

a Find the values of x and y.

b Determine the length of the longest diagonal.



5 Consider the parallelogram $ABCD$ that has  graphed on the number plane.



a Find the slope of sides:

i $\overline{AB}$ and $\overline{CD}$

ii $\overline{AD}$ and $\overline{BC}$

b One property of parallelograms is that opposite sides are parallel. State whether each of the following is a property of parallel lines.

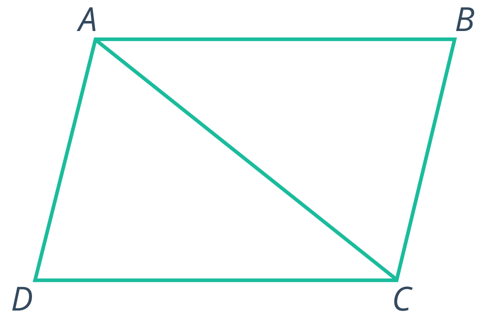
i They intersect.

ii Parallel lines have equal slopes.

iii Parallelograms are the only shapes in which parallel lines appear.

iv Two parallel lines have equal lengths.

6 Consider the following proof to show that the opposite sides of parallelogram $ABCD$ are congruent:



To prove: ?

	Statements	Reasons
1.	$ABCD$ is a parallelogram	Given
2.	$\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3.	$\angle BAC \cong \angle DCA$ and $\angle BCA \cong \angle DAC$	Alternate Interior Angles Theorem
4.	$\overline{AC} \cong \overline{AC}$	Reflexive property of congruence
5.	$\triangle ABC \cong \triangle CDA$	Angle-Side-Angle Congruence
6.	$\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$	Corresponding parts of congruent triangles are congruent (CPCTC)

Determine whether each of the following statements is shown to be true from the given proof.

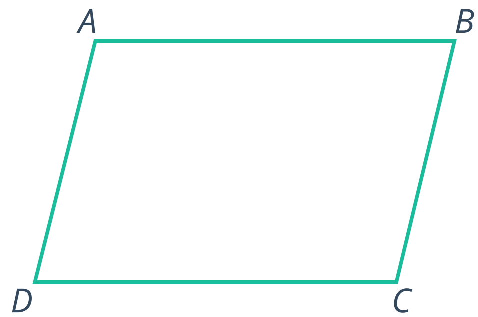
a If a quadrilateral is a parallelogram, then its opposite angles are congruent.

b If a quadrilateral has opposite sides congruent, then it is a parallelogram.

c If a quadrilateral is a parallelogram, then its opposite sides are congruent.

d If a quadrilateral has consecutive interior angles supplementary, then its opposite sides are congruent.

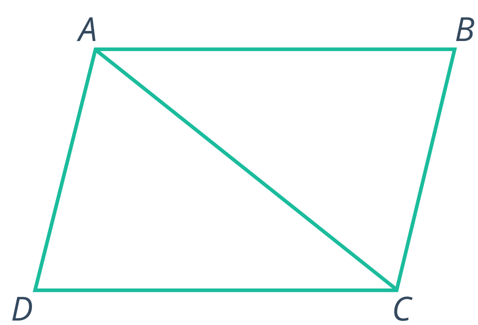
7 Given the parallelogram $ABCD$, write a statement that can be deduced from the proof.



To prove: ?

	Statements	Reasons
1.	$ABCD$ is a parallelogram	Given
2.	$\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3.	$\angle A$ and $\angle B, \angle C$ and $\angle D$, and $\angle D$ and $\angle A$ are supplementary.	Consecutive interior angles postulate

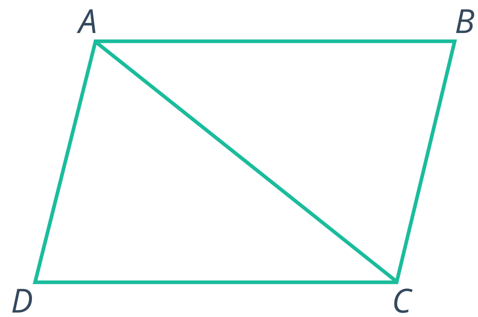
8 Given the parallelogram $ABCD$, complete the following two-column proof to show that $\triangle ABC$ is congruent to $\triangle CDA$:



To prove: $\triangle ABC \cong \triangle CDA$

	Statements	Reasons
1.	$ABCD$ is a parallelogram	Given
2.	$\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3.	$\angle BAC \cong \angle DCA$ and $\angle BCA \cong \angle DAC$	Alternate Interior Angles Theorem
4.	-	-
5.	$\triangle ABC \cong \triangle CDA$	Angle-Side-Angle Congruence

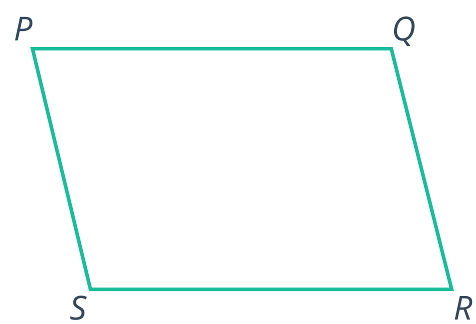
9 Given the parallelogram $ABCD$, complete the following two-column proof to show that the opposite sides are congruent:



To prove: Opposite sides of a parallelogram are congruent

	Statements	Reasons
1.	$ABCD$ is a parallelogram	Given
2.	$\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3.	$\angle BAC \cong \angle DCA$ and $\angle BCA \cong \angle DAC$	Alternate Interior Angles Theorem
4.	$\overline{AC} \cong \overline{AC}$	Reflexive property of congruence
5.	-	-
6.	$\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$	Corresponding parts of congruent triangles are congruent (CPCTC)

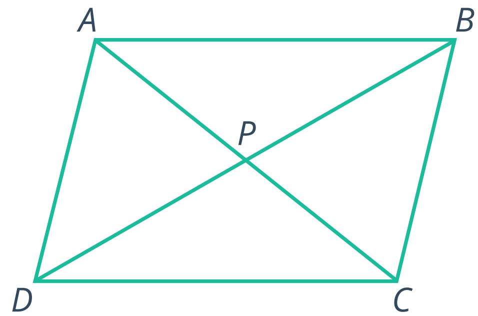
10 Given the parallelogram $PQRS$, complete the following two-column proof to prove that the opposite angles are congruent:



To prove: Opposite angles of a parallelogram are congruent.

	Statements	Reasons
1.	$PQRS$ is a parallelogram	Given
2.	$\overline{PQ} \parallel \overline{SR}$ and $\overline{PS} \parallel \overline{QR}$	Definition of a parallelogram
3.	$\angle P$ and $\angle S, \angle S$ and $\angle R$, and $\angle R$ and $\angle Q$ are supplementary.	Consecutive interior angles postulate
4.	-	-

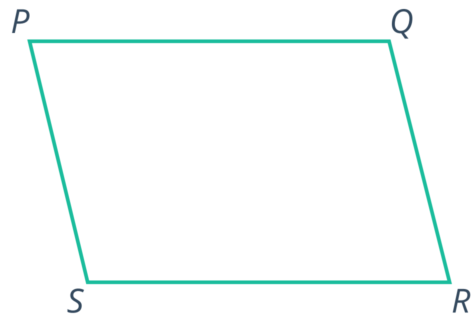
11 Given the parallelogram $ABCD$, complete the following two-column proof to show that the diagonals bisect each other:



To prove: Opposite sides of a parallelogram are congruent

	Statements	Reasons
1.	$ABCD$ is a parallelogram	Given
2.	$\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3.	$\angle ABD \cong \angle CDB$ and $\angle ADB \cong \angle CBD$	Alternate Interior Angles Theorem
4.	$\overline{BD} \cong \overline{BD}$	Reflexive property of congruence
5.	$\triangle ABD \cong \triangle CDB$	Angle-Side-Angle congruence theorem
6.	-	-
7.	-	-
8.	$\triangle ABP \cong \triangle CDP$	Angle-Side-Angle congruence theorem
9.	$\overline{BP} \cong \overline{DP}$ and $\overline{AP} \cong \overline{CP}$	Corresponding parts of congruent triangles are congruent (CPCTC)
10.	$\overline{AC}$ bisects $\overline{BD}$ and $\overline{BD}$ bisects $\overline{AC}$	Definition of a bisector

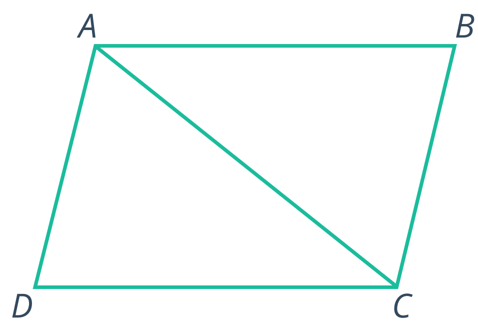
12 Given the parallelogram $PQRS$, complete the following two-column proof to prove that the consecutive angles are supplementary:



To prove: Consecutive angles of a parallelogram are supplementary

	Statements	Reasons
1.	$PQRS$ is a parallelogram	Given
2.	-	-
3.	-	-

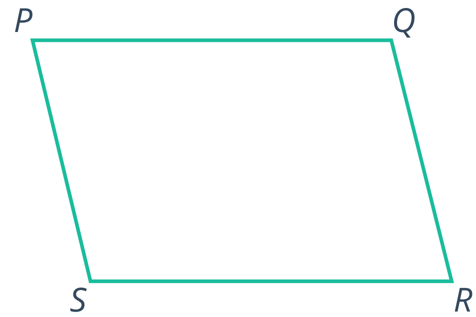
13 Consider the two-colum proof to show that the opposite sides of the given parallelogram $ABCD$ are congruent:



To prove: Opposite sides of a parallelogram are congruent

	Statements	Reasons
1.	$ABCD$ is a parallelogram	Given
2.	$\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
3.	$\angle BAC \cong \angle DCA$ and $\angle ACB \cong \angle CAD$	Alternate Interior Angles Theorem
4.	$\overline{AC} \cong \overline{AC}$	Reflexive property of congruence
5.	$\triangle ABC \cong \triangle CDA$	Angle-Side-Angle Congruence
6.	$\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$	If a quadrilateral is a parallelogram, then it has both pairs of opposide sides congruent

14 Consider the two-colum proof to show that the opposite angles of the given parallelogram $PQRS$ are congruent:



To prove: Opposite angles of a parallelogram are congruent.

	Statements	Reasons
1.	$PQRS$ is a parallelogram	Given
2.	$\angle P$ and $\angle S, \angle S$ and $\angle R,$ and $\angle R$ and $\angle Q$ are supplementary.	Consecutive interior angles postulate
3.	$\overline{PQ} \parallel \overline{SR}$ and $\overline{PS} \parallel \overline{QR}$	Definition of a parallelogram
4.	$\angle P \cong \angle R$ and $\angle Q \cong \angle S$	Congruent supplements theorem

State the error in the two-column proof.

Additional questions

15 Determine whether each statement is true or false:

a A parallelogram has two pairs of opposite sides that are congruent

b The diagonals of a parallelogram bisect the angles

c If the opposite angles of a quadrilateral are congruent, then the quadrilateral is a parallelogram

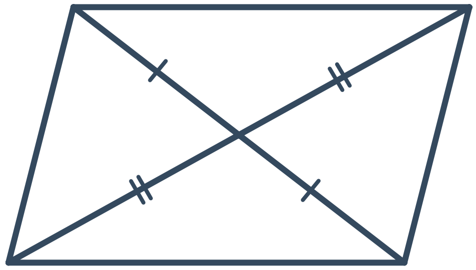
d The diagonals of a parallelogram bisect one another.

e A parallelogram has consecutive supplementary angles.

f A parallelogram has at least one right angle.

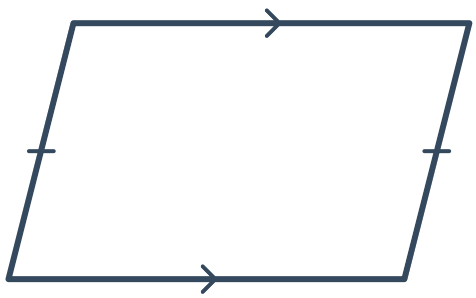
16 Based on the given markings, determine wh[redacted] or not we can conclude the figure is a parallelogram:

a

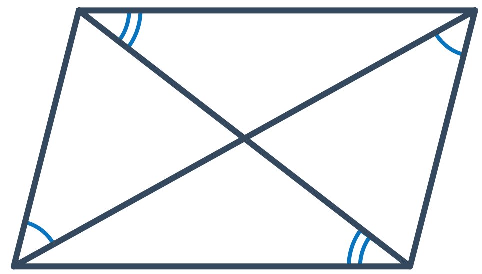


b

c



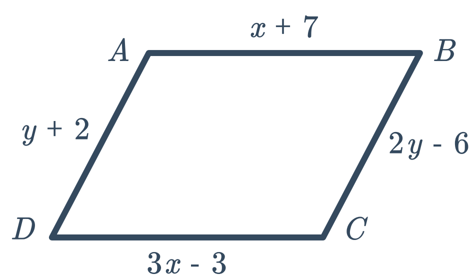
d



17 Given the parallelogram $ABCD$:

a Solve for the variables.

b Solve for the perimeter of $ABCD$.

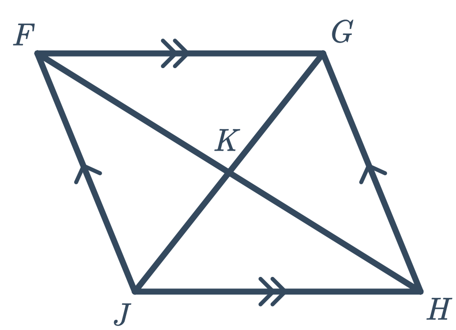


18 Given the parallelogram $FGHJ$:

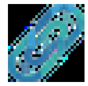
a If $HF = 32$, find KF.

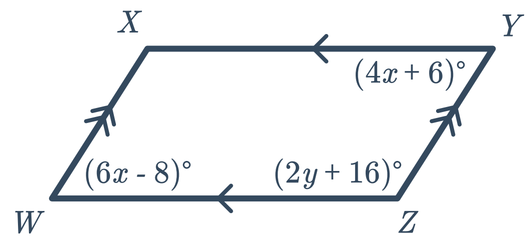
b If $\angle F = 55^\circ$, find $\angle H$.

c If $\angle G = 127^\circ$, find $\angle H$.

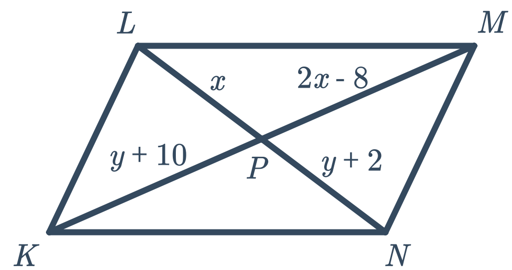


19 Find the value of y that makes $WXYZ$ a parallelogram.





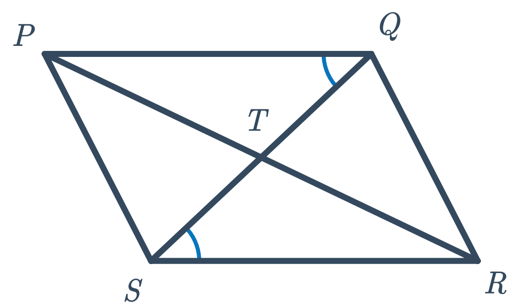
20 Solve for LN in parallelogram $KLMN$.



21

a What additional piece of information about the angles in $PQRS$ would allow you to prove it is a parallelogram?

b What additional piece of information about the sides of $PQRS$ would allow you to prove it is a parallelogram?

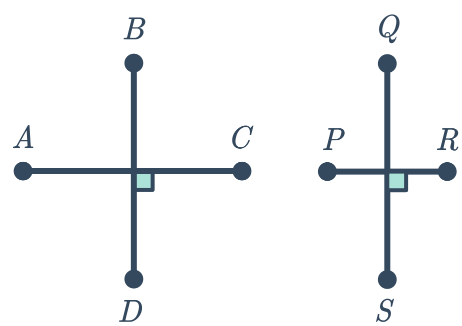


22 Given: $\overline{AC} \perp \overline{BD}$.
$\overline{PR} \perp \overline{QS}$

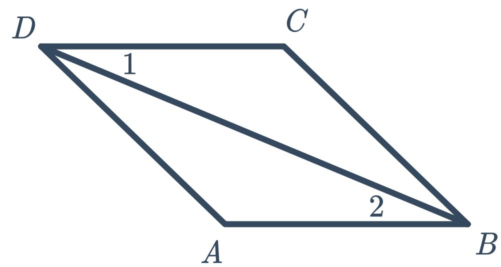
a Copy the diagrams and connect the endpoints in alphabetical order, forming two quadrilaterals.

b Is $ABCD \cong PQRS$? Explain how you know.

c What information would be required for conclude that $ABCD$ and $PQRS$ to be classified as parallelograms?



23 Given: $\overline{AB} \cong \overline{CD}$
$\angle 1 \cong \angle 2$
Prove: $ABCD$ is a parallelogram



24 Rani is constructing a garden in his yard. He plans to build a fence around the garden in the shape of a quadrilateral. Rani has three design options in mind for his garden:

- Option 1: The diagonals of the garden are perpendicular.
- Option 2: One pair of opposite corners form 35° angles, while the other pair of opposite corners form 125° angles.
- Option 3: One pair of opposite sides are 8 ft. long.

Will any of Rani's garden designs form a parallelogram? Explain how you know.

25 Given: Parallelogram $ABCD$
Prove: E is the midpoint of $\overline{AC}$